Question 1 - Speaker Diarization Evaluation and Leaderboard Design

A Methodological Architecture for Conversational Multi-Speaker Diarization in Indian Audio Contexts

1. Purpose and Evaluation Stakeholders

Speaker diarization solves the foundational problem of estimating 'who spoke when' in audio streams, generating time intervals attributed to anonymous speaker IDs. A single aggregate score cannot satisfy diverse industry requirements: meeting intelligence platforms prioritize conversational turn-taking and overlap attribution; contact centers need turn-level compliance and agent-customer segmentation; speech analytics vendors depend on temporal stability for downstream LLM prompts; and enterprise buyers demand empirical trade-offs across accuracy, latency, deployment security, and cost per hour. This benchmark design establishes a reproducible, multi-dimensional evaluation standard specifically calibrated for challenging Indian conversational acoustic environments.

2. Evaluation Shortlist: Ten Representative Systems

The evaluation shortlist spans self-hosted neural diarizers and leading commercial speech APIs. All product capabilities, underlying models, and configurations must be pinned on evaluation day from provider metadata.

System	Provider	Model Checkpoint to Freeze	Deployment	Diarization Mode / Overlap	Official Documentation URL
pyannote.audio	pyannote	speaker-diarization-3.1	Self-hosted	Offline batch; native multi-speaker overlap	https://github.com/pyannote/pyannote-audio
NVIDIA NeMo	NVIDIA	MSDD / Clustering / Sortformer	Self-hosted	Batch & streaming neural diarizer	https://docs.nvidia.com/nemo-framework/user-guide/latest/nemotoolkit/asr/speaker_diarization/models.html
Deepgram	Deepgram	Nova-3 (diarize=true, diarize_model=latest)	Commercial API	Batch & streaming diarization	https://developers.deepgram.com/docs/diarization
AssemblyAI	AssemblyAI	Universal-2 (speaker_labels=true)	Commercial API	Batch transcription + speaker clustering	https://www.assemblyai.com/docs/speech-to-text/speaker-diarization
Google Cloud	Google	Chirp 2 (diarization_config)	Commercial API	Batch & streaming Speech-to-Text	https://cloud.google.com/speech-to-text/docs/multiple-voices
Azure AI Speech	Microsoft	ConversationTranscriber / Batch STT	Commercial API	Real-time & batch conversation diarization	https://learn.microsoft.com/en-us/azure/ai-services/speech-service/get-started-stt-diarization
AWS Transcribe	Amazon	Amazon Transcribe (ShowSpeakerLabels)	Commercial API	Batch & streaming speaker labels	https://docs.aws.amazon.com/transcribe/latest/dg/diarization.html
Speechmatics	Speechmatics	Enhanced model with diarization	Commercial API	Batch & real-time diarization	https://docs.speechmatics.com/features/speaker-diarization
Gladia	Gladia	Gladia STT with speaker diarization	Commercial API	Batch API diarization architecture	https://docs.gladia.io/docs/speaker-diarization
ElevenLabs	ElevenLabs	Scribe v2 with speaker labels	Commercial API	Batch STT with speaker diarization	https://elevenlabs.io/docs/api-reference/speech-to-text

Terminology Disclosure: *AWS Transcribe documentation uses the terms 'speaker labels' or 'speaker identification'. In this benchmark, AWS output is evaluated strictly as anonymous speaker clustering (diarization), NOT as biometric speaker recognition or personal identity verification.*

3. Version Freezing, Capability Disclosure, and Vendor Limitations

Commercial cloud APIs present rolling update risks: vendors frequently release undocumented backend updates without version-tag increments. To enforce rigorous scientific reproducibility: (a) exact API request dates, query parameters, SDK versions, and returned response headers must be logged; (b) request JSON payloads and raw provider responses must be permanently archived; and (c) if an official endpoint fails or mutates during evaluation, the capability is flagged as 'Unknown / Disclosed Limitation' rather than inferred.

4. Committed Primary Scoring Policy & Algorithmic Rules

To prevent ambiguity, the benchmark explicitly commits to a single primary scoring standard:
- **Primary Metric:** Diarization Error Rate (DER) evaluated at **0.0 s collar (zero-collar)** with **overlap included**. Conversational Indian speech features frequent interruptions and short turn durations (median 1.2 s). Forgiving 250 ms at every boundary artificially obscures turn-boundary errors.
- **Secondary / Legacy Track:** DER evaluated with 0.25 s collar (reported in auxiliary columns for historical benchmark comparison).
- **Speaker Mapping Algorithm:** Global Hungarian (Munkres) optimal bipartite matching per session, maximizing total overlapping speech duration between reference and hypothesis speakers. No cross-session speaker tracking is assumed.
- **Speaker-Count Policy:** Unconstrained / autonomous speaker counting. Systems must determine speaker cardinality autonomously. Ground-truth speaker count hints are strictly forbidden on the primary leaderboard; if an API mandates an upper bound, a uniform range of 1-8 is passed to all eligible systems.

5. Mathematical Definitions of Core Diarization Metrics

Diarization Error Rate (DER): Let T_ref be total reference speech time across all speakers (including overlapping speech). DER is the fraction of reference speech incorrectly attributed:

DER = (T_miss + T_fa + T_conf) / T_ref

where *T_miss* (Missed Speech) is reference speech detected as silence; *T_fa* (False Alarm) is hypothesis speech where no reference speech exists; and *T_conf* (Speaker Confusion) is hypothesis speech attributed to the wrong mapped speaker under the optimal Hungarian bijection.

Jaccard Error Rate (JER): Evaluates per-speaker segmentation accuracy with equal weight per speaker regardless of total talking time:

JER = (1 / |S_eval|) * sum_{s in S_eval} [1 - |R_s intersect H_m(s)| / |R_s union H_m(s)|]

where R_s is reference speech for speaker s, H_m(s) is hypothesis speech for mapped speaker m(s), and S_eval contains all reference speakers plus unmapped hypothesis speakers (penalized at 1.0). JER prevents a single dominant talker from masking complete failure on minor conversational participants.

6. Stratified 100-Hour Evaluation Corpus

The evaluation corpus totals exactly **100.0 hours** across ~250 held-out conversational sessions, reflecting realistic Indian linguistic and acoustic diversity:
- **Stratum 1 (40.0 hrs, 100 sessions):** Indian English conversations across professional and casual domains (1-8 speakers; 50% in-person, 50% online).
- **Stratum 2 (25.0 hrs, 65 sessions):** Hindi and Hinglish conversational code-switching (1-6 speakers; mixed in-person and online).
- **Stratum 3 (20.0 hrs, 50 sessions):** Regional Indian languages with English code-switching: Tamil, Telugu, Marathi, and Bengali (2-6 speakers).
- **Stratum 4 (15.0 hrs, 35 sessions):** Acoustic & noise stress cases: 8 kHz PSTN telephony, reverberant halls, cafeteria ambient noise, and low-SNR field recordings (2-8 speakers).

Total allocation: **40.0 + 25.0 + 20.0 + 15.0 = 100.0 hours**. Speaker distribution across all strata: 25% (1-2 spk), 45% (3-4 spk), 25% (5-6 spk), 5% (7-8 spk). Minimum 20% overlapping speech.

7. Hardware Capture Device Allocation & Cross-Tabulation Matrix

Hardware capture device significantly impacts acoustic SNR, reverberation, and far-field microphone array beamforming. The 100 hours are cross-tabulated across physical hardware and acoustic settings:

Acoustic Setting / Strata	Smartphone Tabletop / Handset	Laptop Built-in Array Mic	Meeting Room Boundary Mic	Headset / Lapel Wearable	Total Hours
In-Person Meetings & Huddles	20.0 hrs	0.0 hrs	20.0 hrs	10.0 hrs	50.0 hrs
Online Conferencing (Meet/Zoom)	0.0 hrs	30.0 hrs	0.0 hrs	10.0 hrs	40.0 hrs
Telephony & Ambient Noise Stress	10.0 hrs	0.0 hrs	0.0 hrs	0.0 hrs	10.0 hrs
Total Allocation Across Corpus	**30.0 hrs (30%)**	**30.0 hrs (30%)**	**20.0 hrs (20%)**	**20.0 hrs (20%)**	**100.0 hrs (100%)**

8. Comprehensive Supporting Metrics Suite

To provide holistic evaluation beyond aggregate DER, the platform computes 10 supporting metrics:
1. **Missed Speech Rate (MISS %):** T_miss / T_ref - measures detector under-coverage and voice activity omission.
2. **False Alarm Rate (FA %):** T_fa / T_ref - measures hallucinated speech over background silence or noise.
3. **Speaker Confusion Rate (CONF %):** T_conf / T_ref - measures speaker classification error during detected speech.
4. **Overlap DER (%):** DER computed exclusively on time intervals where >=2 reference speakers speak simultaneously.
5. **Speaker Count Error (SCE):** Mean Absolute Error |N_hyp - N_ref| and signed bias (N_hyp - N_ref) measuring cardinality drift.
6. **Speech Coverage (%):** Hypothesis speech duration overlapping reference speech divided by total reference duration.
7. **Real-Time Factor (RTF):** Total processing wall-clock time divided by audio duration (latency / duration).
8. **Latency / Turnaround (s):** P50 and P90 turnaround time from session ingestion to completed RTTM emission.
9. **Cost per Audio Hour ($/hr):** Commercial API list pricing or dedicated GPU cloud infrastructure amortization cost.
10. **Evaluation Failure Rate (%):** Percentage of sessions triggering 4xx/5xx HTTP errors, API timeouts, or zero-length/empty RTTMs.

9. Annotation Quality Assurance & Statistical Uncertainty

Two-Stage Adjudication QA: Dual independent annotators mark turn boundaries, overlaps, fillers, and laughter. Disagreements exceeding 200 ms at boundaries or any speaker identity dispute trigger formal third-party adjudication. Synthetic validation scripts verify positive durations and absence of overlapping segments per speaker.

Paired Stratified Bootstrap (95% CI): To determine whether ranking differences are statistically significant, whole conversations are resampled with replacement (B=2,000 iterations). Paired delta DER confidence intervals are computed. If the 95% confidence interval for (DER_A - DER_B) spans zero, the systems are declared statistically tied.

10. Public Leaderboard Architecture & Reflection

The public leaderboard features interactive multi-dimensional filtering (Accent, Setting, Speaker Count, Overlap, Track) and model drill-down cards displaying breakdown curves across SNR, conversation duration, and failure rates. **Strategic Reflection:** The most impactful methodological choice is the zero-collar overlap-inclusive policy: while strict, it directly exposes systems that collapse under multi-party conversational Indian audio, providing buyers with an uncompromised measure of operational readiness.